

Question 1

Dataset - [NSGS Infrastructure](#)

Problem: The National "Green Grid" Resilience Audit

Data: [NSGS Infrastructure](#)

Business Context:

The Ministry of Urban Infrastructure is conducting a "National Green Grid Audit" to evaluate the return on environmental investment. Management suspects that high-budget districts are not necessarily the most efficient, and they are concerned that extreme "Power Surge" events (statistical outliers) are distorting the national performance averages.

The Challenge:

Your primary objective is to identify the "Sustainability Leader"—the Mayor whose district achieved the highest "Normalized Energy Yield" relative to their allocated annual budget. To produce a professional, audit-ready report, you must construct an analytical model that independently addresses the following requirements:

- **Data Integrity:** Your model must be mathematically robust; you must neutralize sensory logging errors in the temperature records by substituting all zeros and blanks with the Average value. This "Clean Temperature" is the thermal driver of your final metric.
- **Normalization:** You must apply a statistical adjustment to the power consumption logs to ensure extreme surges do not disproportionately skew the final national baseline by capping outliers at the upper threshold ($Q3 + 1.5 \times IQR$).
- **Feature Engineering:** You must develop a new performance metric, the "Zonal Efficiency Index" (ZEI), defined as the ratio of a district's successfully utilized thermal power relative to its budget:
Formula: $ZEI = \frac{[(Adjusted\ Peak\ Load \times Renewable\ Usage\ \% \times Clean\ Temperature)]}{Annual_Budget_Cr}$
- **Bivariate Diagnosis:** For the "North" region specifically, you must evaluate the relationship between financial investment (Budget) and operational output (ZEI) to determine if a predictable trend exists.

Question:

Based on your final constructed data model, what is the Correlation Coefficient (CORREL) for the North region, which Statistical Trend is visible in the resulting scatter plot, and which Mayor leads the league in average ZEI?

Question 2

Dataset: [I-League](#)

Scenario: The League Board is identifying the "**Best Home Budget Club**" - a team that maintains a low operational budget but remains a high-scoring threat when playing in their own stadium. You are tasked with identifying the top performer among the **three teams with the absolute lowest annual budgets**. Specifically, you must determine which of these three budget-constrained clubs scored the highest cumulative total of "**Home_Goals**".

Question:

After identifying the highest-scoring home team from the low-budget group, determine how many players currently on that specific club's roster are **under the age of 23**.